Supplementary material

Channel attention in lightweight hyperspectral CNNs: an equivalence-tested comparison

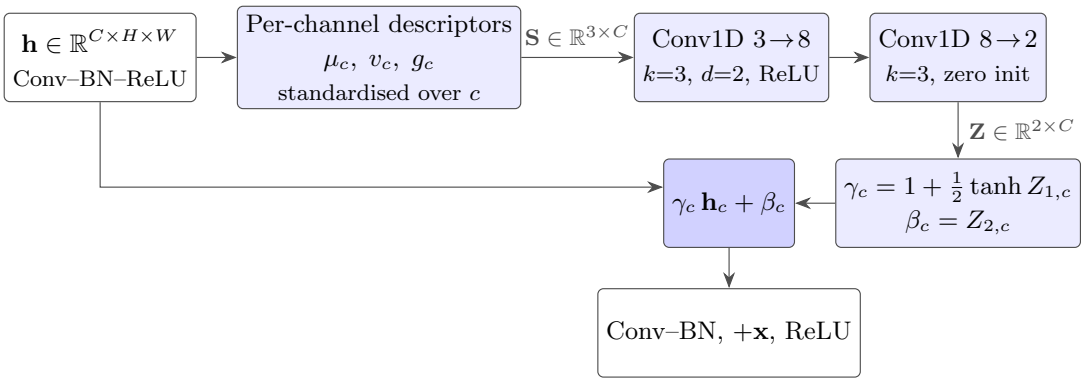


Figure S1: The SNSM probe inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ (mean, variance, cross-channel gradient) are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale γ_c and a shift β_c for each channel. The second convolution starts at zero, so the block is initially the identity.

Table S1: Random 5% splits: OA and AA (%, mean ± std) of all models, including an FC (γ+β) gate and two external baselines trained under the same protocol, HybridSN (original layer configuration) and SSFTT (four tokens, one transformer layer with eight heads). The eight backbone variants use 15 seeds on IP and five elsewhere; the other models use five.

Method	IP		KSC		PU		BW	
	OA	AA	OA	AA	OA	AA	OA	AA
Plain CNN	96.28±0.79	89.49±2.44	99.65±0.34	99.51±0.36	**99.59±0.07**	99.01±0.16	99.09±0.55	98.87±0.90
SE	96.34±0.62	90.04±2.18	99.61±0.28	99.43±0.38	99.53±0.19	98.80±0.56	98.95±0.82	98.76±1.07
ECA	96.28±0.68	90.87±2.53	99.61±0.47	99.40±0.66	99.58±0.10	99.00±0.20	**99.13±0.76**	**98.90±1.08**
CBAM	96.19±0.60	91.58±1.49	99.54±0.56	99.23±0.83	99.58±0.06	**99.11±0.18**	98.99±0.87	98.78±1.20
GCT	96.28±0.80	90.07±2.34	**99.69±0.26**	**99.55±0.28**	99.58±0.08	98.99±0.18	99.11±0.56	98.88±0.91
SRM	96.20±0.71	93.04±2.14	99.69±0.36	99.51±0.49	99.55±0.07	98.92±0.17	98.87±0.63	98.67±0.97
SNSM (γ)	96.17±0.73	92.89±2.00	99.69±0.26	99.50±0.39	99.49±0.05	98.91±0.16	99.10±0.62	98.89±0.96
SNSM (γ+β)	96.27±0.67	**93.33±1.68**	99.63±0.55	99.45±0.73	99.47±0.05	98.93±0.16	99.04±0.69	98.85±1.03
FC (γ+β)	96.24±0.52	92.51±1.60	99.63±0.40	99.48±0.47	99.51±0.06	98.88±0.14	98.80±0.70	98.63±0.98
HybridSN	88.30±4.91	73.64±8.28	95.00±3.02	91.20±5.32	97.84±1.52	95.46±4.12	77.16±37.92	75.56±38.31
SSFTT	**96.71±0.79**	92.35±1.81	99.38±0.28	99.15±0.32	–	–	–	–

Table S2: Paired AA differences SNSM (γ+β) − block (points) with 95% t intervals, for every setting of the main paper's paired table.

Setting	Plain	SE	ECA	CBAM	GCT	SRM	SNSM (γ)	Seeds
IP, random 5%	+3.84 [+2.4, +5.3]	+3.29 [+2.2, +4.4]	+2.45 [+0.9, +4.0]	+1.75 [+0.9, +2.6]	+3.26 [+2.0, +4.6]	+0.29 [-0.8, +1.4]	+0.44 [-0.3, +1.2]	15
IP, smallest 15	+0.39 [+0.1, +0.7]	+0.50 [+0.3, +0.7]	+0.33 [+0.1, +0.6]	+0.60 [+0.3, +0.9]	+0.35 [+0.0, +0.7]	+0.41 [+0.0, +0.8]	+0.30 [+0.1, +0.5]	15
IP, random 50/15	-0.01 [-0.2, +0.2]	-0.08 [-0.2, +0.1]	+0.04 [-0.1, +0.2]	+0.05 [-0.1, +0.2]	+0.01 [-0.2, +0.2]	+0.07 [-0.1, +0.2]	+0.06 [-0.2, +0.3]	15
IP, standard	+0.24 [+0.0, +0.4]	+0.23 [+0.0, +0.4]	+0.33 [+0.2, +0.5]	+0.04 [-0.2, +0.2]	+0.25 [+0.1, +0.4]	-0.02 [-0.2, +0.2]	+0.12 [-0.1, +0.3]	15
PU, smallest 1/1/2	-0.59 [-1.6, +0.4]	-0.74 [-1.6, +0.1]	-1.19 [-1.9, -0.5]	-0.30 [-1.1, +0.5]	-1.14 [-2.2, -0.1]	-0.55 [-1.2, +0.2]	-1.59 [-2.6, -0.5]	15
KSC, smallest 1/1/2	-0.24 [-0.9, +0.5]	+0.79 [+0.1, +1.4]	+0.39 [-0.6, +1.4]	+0.54 [-0.7, +1.8]	-0.21 [-1.0, +0.5]	+1.41 [+0.3, +2.5]	+0.03 [-1.3, +1.3]	15
PU, hardest 1/1/2	+0.97 [-0.1, +2.1]	+0.98 [-0.2, +2.1]	+0.64 [-0.5, +1.8]	+0.96 [-0.4, +2.3]	+0.77 [-0.2, +1.8]	+0.79 [-0.3, +1.9]	-0.82 [-1.5, -0.1]	15
KSC, hardest 1/1/2	+0.71 [-0.0, +1.4]	+0.88 [-0.1, +1.8]	+0.20 [-0.5, +0.9]	+1.14 [+0.3, +2.0]	+0.77 [+0.1, +1.4]	+0.66 [-0.3, +1.6]	+0.90 [-0.3, +2.1]	15
WHU, random 5%	-0.06 [-0.1, +0.0]	-0.01 [-0.1, +0.1]	-0.02 [-0.1, +0.0]	+0.00 [-0.1, +0.1]	+0.00 [-0.1, +0.1]	-0.06 [-0.2, +0.0]	-0.03 [-0.1, +0.1]	10
PU, patch-disjoint	+0.31 [-1.9, +2.5]	-0.47 [-2.7, +1.8]	-0.39 [-2.4, +1.7]	+1.03 [-0.7, +2.8]	+0.92 [-1.0, +2.8]	+1.33 [-0.4, +3.1]	+0.03 [-2.0, +2.0]	15
KSC, patch-disjoint	+3.46 [+1.5, +5.4]	-0.19 [-2.3, +1.9]	+1.86 [+0.3, +3.4]	+0.91 [-0.9, +2.7]	+3.51 [+1.5, +5.5]	+4.25 [+1.7, +6.8]	+1.40 [-1.0, +3.8]	15
WHU, patch-disjoint	-2.01 [-3.3, -0.7]	-2.39 [-4.8, +0.1]	-1.77 [-4.5, +1.0]	-1.85 [-3.6, -0.1]	-1.32 [-3.3, +0.6]	-0.84 [-2.7, +1.0]	-0.46 [-3.0, +2.1]	10
IP, random 5%, deeper host	-0.53 [-1.4, +0.3]	+0.14 [-0.6, +0.9]	+0.18 [-0.5, +0.9]	-0.35 [-1.2, +0.5]	-0.51 [-1.3, +0.2]	-0.87 [-2.2, +0.5]	-0.19 [-1.3, +0.9]	10
KSC, patch-disjoint, deeper host	-1.88 [-4.9, +1.1]	-0.80 [-5.5, +3.9]	-2.87 [-5.2, -0.6]	-1.35 [-4.7, +2.0]	-1.63 [-4.6, +1.3]	+0.34 [-2.8, +3.4]	-2.00 [-5.6, +1.6]	10
KSC, patch-disjoint, train-only PCA	+0.33 [-2.3, +2.9]	-0.98 [-3.1, +1.2]	-0.01 [-2.4, +2.4]	-2.38 [-4.1, -0.7]	+0.54 [-2.0, +3.1]	+1.65 [-0.7, +4.0]	-0.38 [-2.8, +2.1]	15
IP, random 5%, lr 3×10^{-4}	+8.16 [+6.8, +9.5]	+7.42 [+4.8, +10.0]	+8.26 [+5.4, +11.2]	+6.12 [+3.5, +8.8]	+8.06 [+6.7, +9.4]	+5.37 [+4.1, +6.6]	+5.27 [+3.1, +7.4]	5
IP, random 5%, lr 3×10^{-3}	+0.98 [-0.0, +2.0]	+1.48 [+0.2, +2.8]	+0.56 [-0.3, +1.4]	+0.80 [-0.3, +1.9]	+0.83 [+0.0, +1.7]	+0.28 [-0.8, +1.3]	-0.03 [-1.2, +1.2]	15

Table S3: Paired AA differences block − plain CNN (points), so that no block serves as the reference. *: Holm-corrected Wilcoxon $p < 0.05$ within the row (seven comparisons); †: equivalent within ±1 point (paired TOST). GCT stays close to the plain CNN in most settings because its gate starts at the identity.

Setting	SE	ECA	CBAM	GCT	SRM	SNSM (γ)	SNSM (γ+β)
IP, random 5%	+0.55	+1.38	+2.09*	+0.58*†	+3.55*	+3.40*	+3.84*
IP, smallest 15	-0.11†	+0.06†	-0.20†	+0.04†	-0.02†	+0.09†	+0.39*†
IP, random 50/15	+0.07†	-0.05†	-0.06†	-0.01†	-0.08†	-0.07†	-0.01†
IP, standard	+0.01†	-0.09†	+0.21†	-0.01†	+0.26*†	+0.13†	+0.24†
PU, smallest 1/1/2	+0.15†	+0.60	-0.29	+0.55†	-0.05†	+0.99	-0.59
KSC, smallest 1/1/2	-1.03*	-0.63	-0.78	-0.03†	-1.65*	-0.27	-0.24†
PU, hardest 1/1/2	-0.01†	+0.33	+0.01†	+0.20†	+0.18†	+1.79*	+0.97
KSC, hardest 1/1/2	-0.17†	+0.52	-0.43	-0.06†	+0.06†	-0.19	+0.71
WHU, random 5%	-0.05†	-0.03†	-0.06†	-0.06†	+0.00†	-0.03†	-0.06†
PU, patch-disjoint	+0.78	+0.70	-0.72	-0.61	-1.02	+0.28	+0.31
KSC, patch-disjoint	+3.65*	+1.60	+2.55	-0.06†	-0.79	+2.06	+3.46*
WHU, patch-disjoint	+0.37	-0.24	-0.16	-0.69	-1.17	-1.55	-2.01
IP, random 5%, deeper host	-0.67	-0.70	-0.18†	-0.01†	+0.34	-0.33	-0.53
KSC, patch-disjoint, deeper host	-1.08	+0.98	-0.54	-0.25	-2.22	+0.12	-1.88
KSC, patch-disjoint, train-only PCA	+1.31	+0.35	+2.71	-0.20†	-1.32	+0.72	+0.33
IP, random 5%, lr 3×10^{-4}	+0.74	-0.10	+2.04	+0.10†	+2.79	+2.89	+8.16
IP, random 5%, lr 3×10^{-3}	-0.50	+0.42	+0.18	+0.14†	+0.70	+1.01	+0.98

Table S4: Sensitivity to the equivalence margin: number of the seven comparisons SNSM (γ+β) − block per setting that are equivalent (paired TOST, $\alpha = 0.05$) within ±0.5, ±1 (main paper) and ±2 AA points.

Setting	±0.5	±1	±2
IP, random 5%	0	0	2
IP, smallest 15	1	7	7
IP, random 50/15	7	7	7
IP, standard	7	7	7
PU, smallest 1/1/2	0	1	5
KSC, smallest 1/1/2	0	2	6
PU, hardest 1/1/2	0	0	6
KSC, hardest 1/1/2	0	1	7
WHU, random 5%	7	7	7
PU, patch-disjoint	0	0	1
KSC, patch-disjoint	0	0	1
WHU, patch-disjoint	0	0	0
IP, random 5%, deeper host	0	2	7
KSC, patch-disjoint, deeper host	0	0	0
KSC, patch-disjoint, train-only PCA	0	0	1
IP, random 5%, lr 3×10^{-4}	0	0	0
IP, random 5%, lr 3×10^{-3}	0	0	6

Table S5: Paired OA differences SNSM (γ+β) − block (points). *: Holm-corrected Wilcoxon $p < 0.05$ within the row; †: equivalent within ±1 point (paired TOST).

Setting	Plain	SE	ECA	CBAM	GCT	SRM	SNSM (γ)	Seeds
IP, random 5%	-0.01†	-0.06†	-0.01†	+0.08†	-0.00†	+0.07†	+0.11†	15
IP, smallest 15	-0.14†	-0.12†	-0.11†	+0.04†	-0.12†	+0.02†	+0.09†	15
IP, random 50/15	+0.04†	-0.15†	+0.09†	+0.12†	+0.06†	+0.10†	+0.01†	15
IP, standard	+0.08†	+0.34†	+0.31†	+0.14†	+0.15†	+0.03†	+0.16†	15
PU, smallest 1/1/2	-0.25†	-0.26†	-0.39*†	-0.10†	-0.39†	-0.18†	-0.47*†	15
KSC, smallest 1/1/2	-0.20†	+0.23†	+0.17†	+0.19†	-0.19†	+0.50†	-0.03†	15
PU, hardest 1/1/2	+0.53	+0.58	+0.31†	+0.50	+0.40†	+0.50	-0.55†	15
KSC, hardest 1/1/2	+0.75	+0.68*	+0.29†	+0.95*	+0.80	+0.56	+0.43	15
WHU, random 5%	-0.04†	-0.01†	-0.01†	-0.02†	-0.03†	-0.05†	-0.02†	10
PU, patch-disjoint	+1.50	+0.89	+0.51	+0.69	+2.21	+0.79	+1.39	15
KSC, patch-disjoint	+1.98	+0.32	+0.72	+0.43	+1.91	+2.75	+0.25	15
WHU, patch-disjoint	-2.51	-2.86	-1.22	-1.22	-0.24	-2.61	+0.07	10
IP, random 5%, deeper host	-0.41†	-0.26†	-0.26†	-0.34†	-0.39†	-0.27†	-0.18†	10
KSC, patch-disjoint, deeper host	-1.12	-0.79	-2.40	-1.66	-1.23	+1.66	-2.13	10
KSC, patch-disjoint, train-only PCA	-0.74	-0.65	+0.23	-2.17	-0.67	-0.22	-0.78	15
IP, random 5%, lr 3×10^{-4}	+0.45†	+0.29†	+0.29†	-0.06†	+0.46†	+0.27†	+0.18†	5
IP, random 5%, lr 3×10^{-3}	-0.23†	-0.03†	-0.23†	-0.26†	-0.22†	-0.12†	-0.07†	15

Table S6: AA (%, mean ± std) of random 5% splits and of the same splits with only the three smallest classes changed (15 seeds; the unchanged PU and KSC splits use five). The last row is the range of the block means.

Block	IP, 5%	IP, smallest 15	PU, 5%	PU, smallest 1/1/2	KSC, 5%	KSC, smallest 1/1/2
Plain CNN	89.49±2.44	96.03±0.76	99.01±0.16	84.20±3.43	99.51±0.36	**93.12±3.44**
SE	90.04±2.18	95.93±0.64	98.80±0.56	84.35±3.29	99.43±0.38	92.09±3.25
ECA	90.87±2.53	96.10±0.76	99.00±0.20	84.80±3.23	99.40±0.66	92.48±3.19
CBAM	91.58±1.49	95.83±0.67	**99.11±0.18**	83.91±2.11	99.23±0.83	92.34±4.15
GCT	90.07±2.34	96.07±0.75	98.99±0.18	84.74±3.75	**99.55±0.28**	93.09±3.24
SRM	93.04±2.14	96.01±0.81	98.92±0.17	84.15±3.45	99.51±0.49	91.47±3.62
SNSM (γ)	92.89±2.00	96.13±0.72	98.91±0.16	**85.19±3.48**	99.50±0.39	92.84±3.64
SNSM (γ+β)	**93.33±1.68**	**96.43±0.66**	98.93±0.16	83.60±2.96	99.45±0.73	92.88±3.12
Range of block means	3.84	0.60	0.31	1.59	0.32	1.65

Table S7: Robustness checks: AA (%, mean ± std) of the eight backbone variants in a deeper, wider host (four residual blocks of 128 channels; ten seeds), with PCA fitted on training pixels only (15 seeds), with the three hardest classes reduced to 1, 1 and 2 training pixels (15 seeds), and with learning rates 3×10^{-4} (five seeds) and 3×10^{-3} (15 seeds). The last row is the range of the block means.

Block	IP, deeper	KSC disj., deeper	KSC disj., train PCA	PU, hardest	KSC, hardest	IP, lr 3e-4	IP, lr 3e-3
Plain CNN	93.85±1.69	60.46±5.77	64.26±4.88	73.19±2.03	88.17±1.35	80.23±1.51	92.15±2.12
SE	93.18±2.02	59.38±6.05	65.57±5.03	73.17±2.09	88.01±1.39	80.97±1.99	91.65±1.84
ECA	93.14±1.80	**61.44±6.66**	64.61±4.56	73.52±2.28	88.69±1.92	80.13±2.35	92.57±1.93
CBAM	93.67±1.95	59.92±6.28	**66.97±6.19**	73.20±2.22	87.74±2.14	82.27±1.98	92.33±1.75
GCT	93.83±1.61	60.21±6.45	64.06±5.14	73.39±2.05	88.11±1.45	80.33±1.43	92.30±2.04
SRM	**94.19±1.91**	58.24±5.28	62.94±5.31	73.37±2.56	88.23±2.26	83.02±1.64	92.85±1.75
SNSM (γ)	93.51±1.59	60.58±5.41	64.98±7.02	**74.98±2.65**	87.98±2.44	83.12±1.49	**93.16±1.89**
SNSM (γ+β)	93.32±2.21	58.58±7.23	64.59±6.63	74.16±2.58	**88.89±1.34**	**88.39±0.78**	93.13±2.17
Range of block means	1.05	3.20	4.03	1.81	1.14	8.26	1.51

Table S8: SNSM (γ+β) probe ablation on IP, random 5% splits (AA, %; five seeds). ΔAA is the paired difference from the full block; wins count seeds on which the variant is better.

Descriptors	AA	ΔAA (wins/5)
mean, var, grad (SNSM)	94.16±1.07	–
mean	93.01±0.72	-1.16 (1/5)
mean, var	94.55±1.67	+0.39 (4/5)
mean, grad	93.37±1.09	-0.80 (1/5)
mean, var, grad; default init.	92.67±1.15	-1.49 (0/5)

Table S9: Blocks applied to the raw bands in wavelength order, without PCA or wavelet features: AA (%, mean ± std, random 5% splits, five seeds). CBAM and SRM were not run in this setting.

Block	IP	PU	KSC
Plain CNN	82.53±0.95	99.00±0.31	99.18±0.43
SE	83.54±2.01	98.91±0.36	98.61±0.89
ECA	82.95±0.71	**99.10±0.22**	**99.28±0.30**
GCT	82.64±0.87	99.07±0.23	99.18±0.60
SNSM (γ)	83.56±1.08	99.10±0.21	98.96±0.86
SNSM (γ+β)	**87.45±2.48**	98.87±0.25	98.83±0.89

Table S10: Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs, random 5% splits). MMACs: multiply–accumulates of convolution and linear layers counted by `thop` (attention products and descriptor statistics excluded). Latency: batch size 1 on a Tesla T4.

Method	Params (M)	MMACs	Latency (ms)	Train IP (s)	Train PU (s)
Plain CNN	0.184	41.2	0.69	4.2	17.6
SE	0.188	41.2	1.17	4.5	19.3
ECA	0.184	41.2	0.98	4.8	19.8
CBAM	0.185	41.3	1.55	5.9	24.9
GCT	0.184	41.2	1.11	4.7	20.0
SRM	0.184	41.2	1.22	4.8	20.2
SNSM (γ)	0.184	41.3	2.54	7.1	29.5
SNSM (γ+β)	0.184	41.3	2.54	7.2	30.2
FC (γ+β)	0.190	41.2	1.31	4.7	19.8
HybridSN	1.743	137.4	0.71	26.2	115.3
SSFTT	0.291	34.9	1.25	7.6	–